

# RSV VACCINE ABRYSVO FROM PFIZER AND ITS METHOD IMPLEMENTATION

Flore Laurent  
OMCL meeting – Ohrid  
May 2024

# Agenda

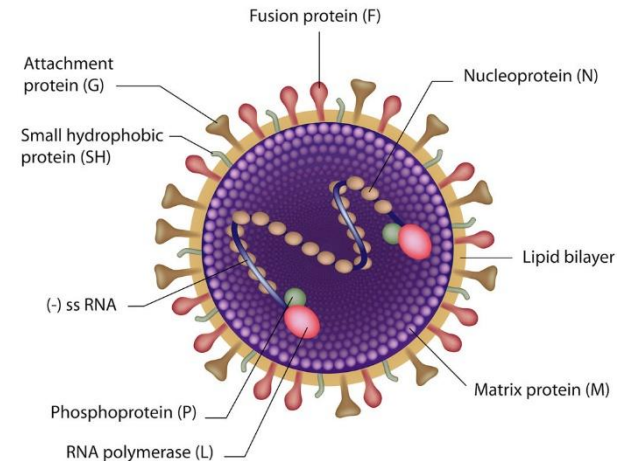
- Product presentation
- Manufacturing process
- Control strategy
- Implementation of method at OMCL
  - *in vitro* relative potency by ELISA
  - Identity by slot blot
  - Purity/Trimer by SE-HPLC
- Challenges/lessons learned

# Product presentation

## ➤ RSV (Respiratory Syncytial Virus)

- Common respiratory virus that usually causes mild, cold-like symptoms.
- Can be serious in vulnerable people (older adults, babies and young, etc.)
- children 64 million sickens & 160.000 deaths per year worldwide
- Seasonal disease. Typically, Oct-May in Northern hemisphere

Respiratory Syncytial Virus



*respiratory-syncytial-virus-(RSV) - Pediatric Pulmonologists (pedilung.com)*

## ➤ Abrysvo vaccine manufactured by Pfizer, dedicated for

- **Older adults** (60 years and older)
- **Pregnant women** (prevent RSV in babies from birth through 6 months of age)

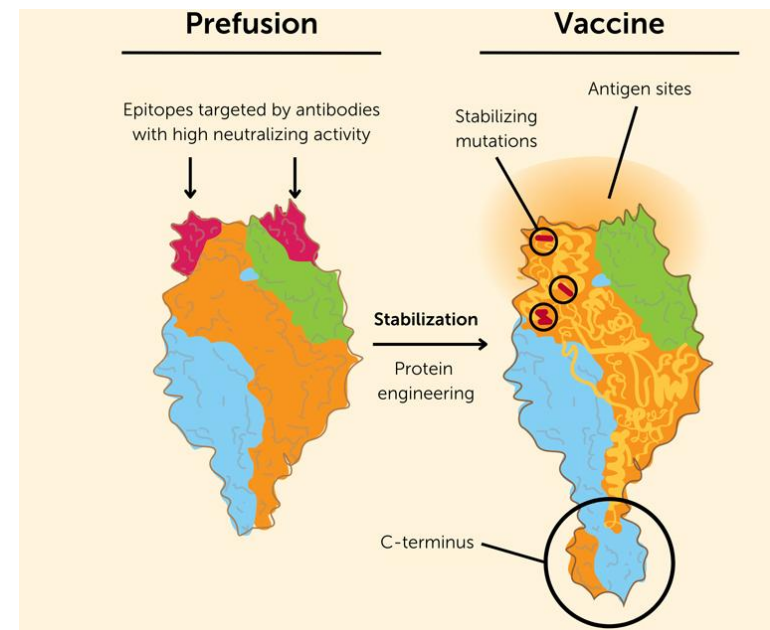
# Product presentation

## ➤ Recombinant, bivalent vaccine ABRYSVO :

- glycoprotein F stabilized in the prefusion conformation
- produced in Chinese Hamster Ovary cells by recombinant DNA technology.
- RSV Subgroup A & B

## ➤ F (fusion) protein facilitates viral entry

- major surface antigen of the RSV
- Elicits major neutralizing Ab activity
- pre and post-fusions forms (prefusion form chosen as vaccine)



*RSV vaccines: the latest success story - IAVI*

# Product presentation

➤ Single dose (0,5 mL) – lyophylised

➤ Composition :

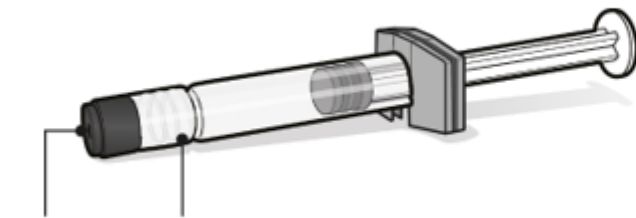


| Ingredient                                                                                                                                                                                                                                                                                                                                          | Grade/Quality Standard | Function          | Composition Amount of Bulk Drug Product (mg/vial) <sup>a</sup> | Composition Amount of Reconstituted Drug Product (mg/mL) | Composition Amount of Dose (mg/dose) <sup>b</sup> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|----------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------|
| 847A Drug Substance                                                                                                                                                                                                                                                                                                                                 | In-house specification | Active Ingredient | 0.08                                                           | 0.12                                                     | 0.06                                              |
| 847B Drug Substance                                                                                                                                                                                                                                                                                                                                 | In-house specification | Active Ingredient | 0.08                                                           | 0.12                                                     | 0.06                                              |
| Tromethamine (Tris base)                                                                                                                                                                                                                                                                                                                            | USP-NF, Ph. Eur., JP   | Buffer component  | 0.15                                                           | 0.23                                                     | 0.11                                              |
| Tris (hydroxymethyl) aminomethane hydrochloride (Tris-HCl)                                                                                                                                                                                                                                                                                          | In-house specification | Buffer component  | 1.41                                                           | 2.07                                                     | 1.04                                              |
| Sucrose                                                                                                                                                                                                                                                                                                                                             | USP-NF, Ph. Eur., JP   | Cryoprotectant    | 15.3                                                           | 22.5                                                     | 11.25                                             |
| Mannitol                                                                                                                                                                                                                                                                                                                                            | USP-NF, Ph. Eur., JP   | Bulking agent     | 30.6                                                           | 45.0                                                     | 22.50                                             |
| Polysorbate 80                                                                                                                                                                                                                                                                                                                                      | USP-NF, Ph. Eur., JP   | Surfactant        | 0.10                                                           | 0.15                                                     | 0.08                                              |
| Sodium Chloride                                                                                                                                                                                                                                                                                                                                     | USP-NF, Ph. Eur., JP   | Tonicifier        | 1.49                                                           | 2.19                                                     | 1.1                                               |
| Water for Injection                                                                                                                                                                                                                                                                                                                                 | USP-NF, Ph. Eur., JP   | Solvent           | Composition Amount (mg/vial) <sup>a</sup>                      | q.s to 1 mL                                              | q.s to 0.5 mL                                     |
| a. Based on a target fill volume of 0.51 mL of bulk drug product per vial for lyophilization<br>b. Each vial contains 0.68 mL of solution after reconstitution with sterile water diluent and vial adapter which will then be withdrawn in the same prefilled syringe of sterile water diluent to enable deliverable volume of not less than 0.5 mL |                        |                   |                                                                |                                                          |                                                   |

# Product presentation

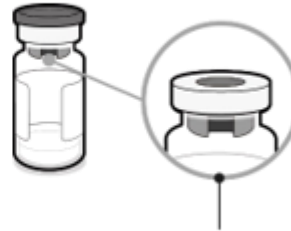
- Powder and solvent for injection (water for injection)
- Glass vial (2 mL container) + pre-filled syringe with solvent

Pre-filled syringe containing solvent for Abrysvo



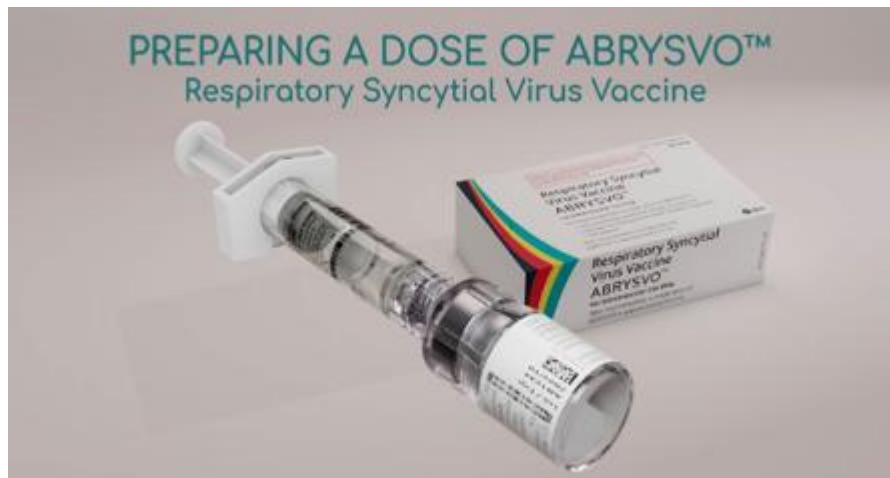
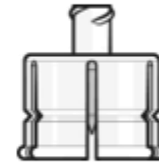
Syringe cap Luer lock adaptor

Vial containing antigens for Abrysvo (powder)



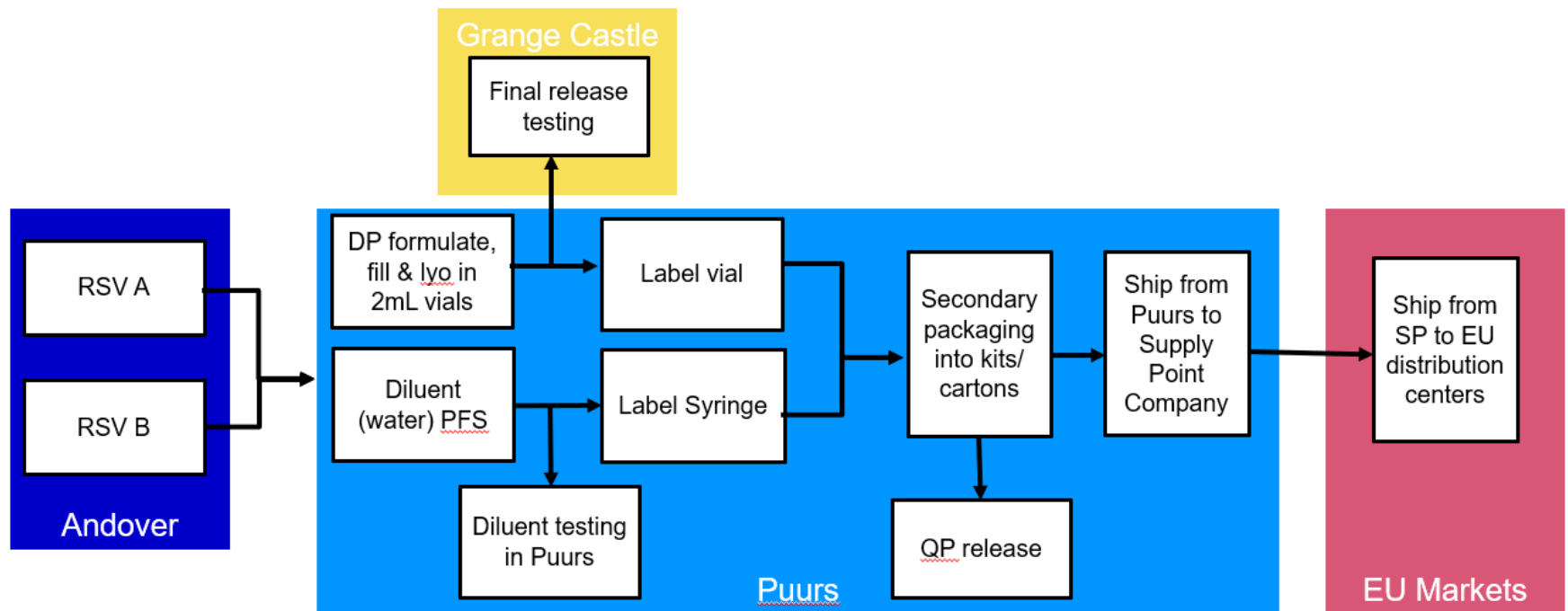
Vial stopper (with flip off cap removed)

Vial adaptor



# Manufacturing process

## RSV EU Supply Chain Map



# Control strategy – Exemple DS 847A

**Table 3.2.S.4.1-1. 847A Drug Substance Specification**

| Quality Attribute                             | Analytical Procedure                | Acceptance Criteria                                                      |
|-----------------------------------------------|-------------------------------------|--------------------------------------------------------------------------|
| Clarity                                       | Appearance (Clarity), compendial    | ≤18 NTU                                                                  |
| Coloration                                    | Appearance (Coloration), compendial | Not more intensely colored than level 5 of the brown (B) color standard. |
| pH                                            | Potentiometry, compendial           | 6.8-7.7                                                                  |
| Protein concentration                         | UV spectroscopy                     | 4.00-5.50 mg/mL                                                          |
| Identity <sup>a</sup>                         | Slot blot                           | Positive                                                                 |
| Relative prefusion content (Potency)          | Prefusion F ELISA                   | 72-129%                                                                  |
| Trimer                                        | Size exclusion HPLC (SE-HPLC)       | ≥94.0%                                                                   |
| High molecular mass species (HMMS)            | Size exclusion HPLC (SE-HPLC)       | ≤2.5%                                                                    |
| Low molecular mass species (LMMS)             | Size exclusion HPLC (SE-HPLC)       | ≤4.5%                                                                    |
| Purity (subunit F1+F2)                        | CGE (reducing) (rCGE)               | ≥94.0%                                                                   |
| Residual host cell protein (HCP) <sup>a</sup> | HCP ELISA                           | ≤1000 ng/mg (0.1% w/w)                                                   |
| Bioburden <sup>b</sup>                        | Bioburden, compendial               | ≤3 CFU/30 mL                                                             |
| Endotoxin <sup>b</sup>                        | Endotoxin (LAL), compendial         | ≤15.6 EU/mg                                                              |

a. The test is not performed on stability.

b. The test is performed at release and the end of shelf life.

Abbreviation: NTU = nephelometric turbidity units; UV = ultraviolet; ELISA = enzyme-linked immunosorbent assay; HPLC = high performance liquid chromatography; CGE = capillary gel electrophoresis; CFU = colony forming units; EU = endotoxin units; LAL= limulus amebocyte lysate.



# Control strategy - DP

**Table 3.2.P.5.1-1. Drug Product Specification**

| Quality Attribute                        | Analytical Procedure                                               | Acceptance Criteria                                                     |
|------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------|
| Appearance (before reconstitution)       | Appearance before reconstitution (visual)                          | White cake essentially free from visible foreign particulates           |
| Residual moisture                        | Karl Fischer, compendial                                           | ≤2.0%                                                                   |
| Reconstitution time                      | Reconstitution time                                                | ≤120 seconds                                                            |
| Clarity                                  | Appearance after reconstitution (clarity), compendial              | ≤6 NTU                                                                  |
| Coloration                               | Appearance after reconstitution (coloration), compendial           | Not more intensely colored than level 7 of the brown (B) color standard |
| Visible particulates                     | Appearance after reconstitution (visible particulates), compendial | Essentially free from visible particulates                              |
| Subvisible particles                     | Particulate matter in injections, compendial                       | Meets compendial requirements                                           |
| pH                                       | Potentiometry, compendial                                          | 6.9-7.7                                                                 |
| Osmolality <sup>a</sup>                  | Osmometry, compendial                                              | 364-492 mOsm/kg                                                         |
| Protein concentration                    | UV spectroscopy                                                    | 216-264 µg/mL                                                           |
| 847A content                             | Peptide mapping                                                    | 43.0-57.0%                                                              |
| 847B content                             | Peptide mapping                                                    | 43.0-57.0%                                                              |
| Uniformity of dosage units <sup>a</sup>  | Uniformity of dosage units, compendial                             | Meets compendial requirements                                           |
| PS80 concentration <sup>a</sup>          | Mixed-mode HPLC-ELSD                                               | 0.09-0.21 mg/mL                                                         |
| Identity <sup>a</sup>                    | Slot Blot                                                          | Positive for 847A and 847B                                              |
| Relative prefusion content (Potency)     | Prefusion F ELISA                                                  | 67-120%                                                                 |
| Trimer                                   | Size exclusion HPLC (SE-HPLC)                                      | ≥90.8%                                                                  |
| High molecular mass species (HMMS)       | Size exclusion HPLC (SE-HPLC)                                      | ≤6.5%                                                                   |
| Low molecular mass species (LMMS)        | Size exclusion HPLC (SE-HPLC)                                      | ≤4.5%                                                                   |
| Endotoxin <sup>c</sup>                   | Endotoxin (LAL), compendial                                        | ≤12.5 EU/mL                                                             |
| Sterility <sup>d</sup>                   | Sterility, compendial                                              | No growth detected                                                      |
| Container closure integrity <sup>b</sup> | Dye ingress                                                        | Pass                                                                    |

Tests performed by OMCL

a. The test is not performed on stability samples.

b. Test not performed at release; only performed annually on stability.

c. Test performed at release and the end of shelf life. Release testing uses the gel clot method, and stability testing uses the kinetic chromogenic method.

d. Test performed at release and the end of shelf life. Rapid sterility test, which is performed in accordance with the compendia with the exception of incubation duration and detection method (see [Section 3.2.P.5.2 Sterility](#)), may also be used.

Abbreviation: NTU = nephelometric turbidity unit; UV= ultraviolet; ELSD = evaporative light-scattering detection; ELISA= enzyme linked immunosorbent assay; HPLC= high performance liquid chromatography; LAL = limulus amoebocyte lysate; EU= endotoxin units.

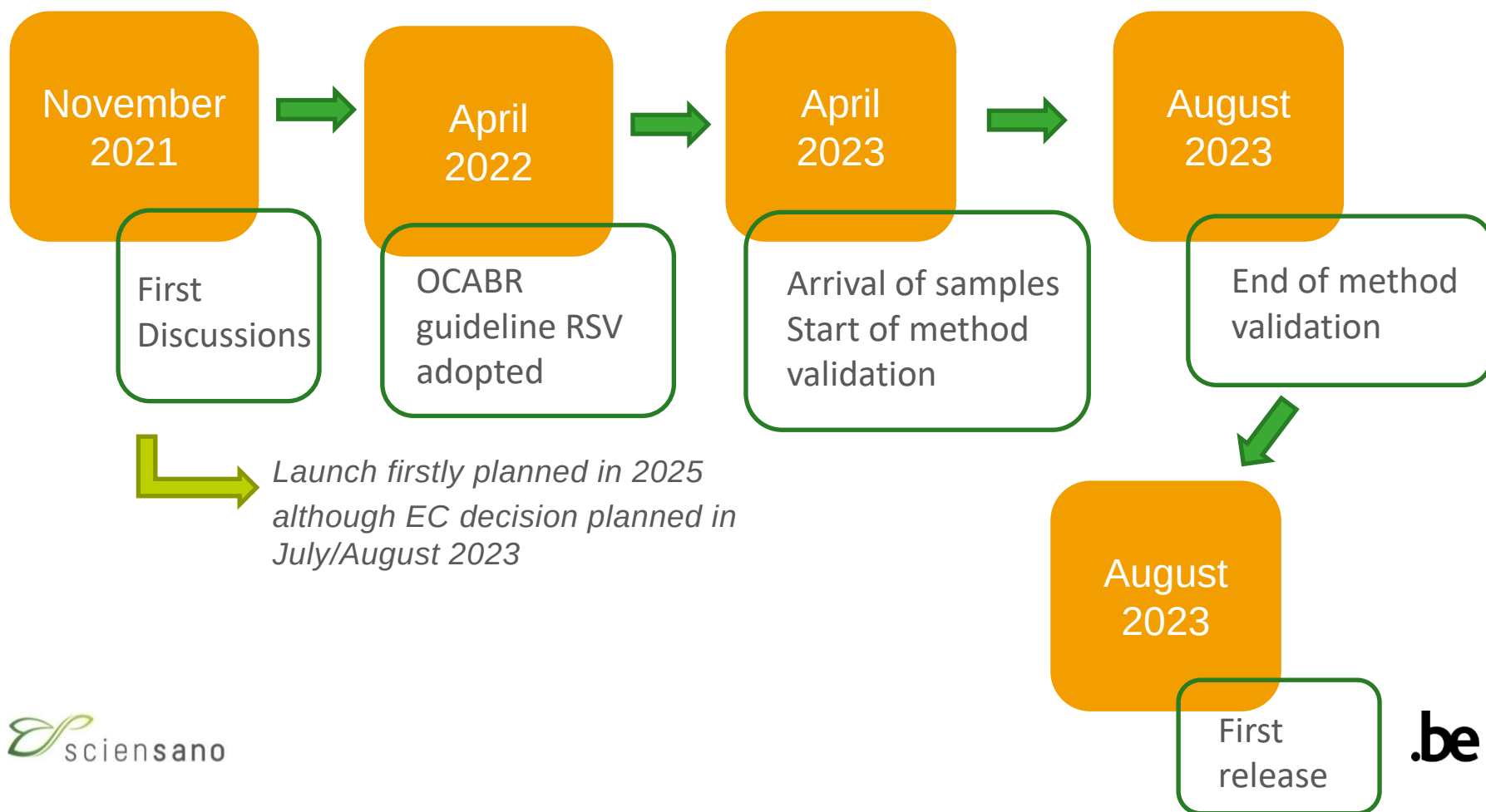
# METHOD IMPLEMENTATION AT SCIENSANO

# Tests performed at OMCL on DP

| Tests                                                    | Manufacturer                                                                                                                                                                                                                                                   | OMCL-B                                                                                                                                                                                                                                                         |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance                                               | White cake essentially free from visible foreign particulates before reconstitution.<br>Colourless to slightly brown solution, not more opalescent than suspension II and not more coloured than solution B7, free from visible particles after reconstitution | White cake essentially free from visible foreign particulates before reconstitution.<br>Colourless to slightly brown solution, not more opalescent than suspension II and not more coloured than solution B7, free from visible particles after reconstitution |
| potency assay                                            | <i>In vitro</i> relative potency :<br>67 - 120 %                                                                                                                                                                                                               | <i>In vitro</i> relative potency :<br>67 - 120 %                                                                                                                                                                                                               |
| Identity by slot blot                                    | Positive for 847A and 847B                                                                                                                                                                                                                                     | Positive for 847A and 847B                                                                                                                                                                                                                                     |
| Trimer percentage by SE-HPLC                             | ≥ 90.8 %                                                                                                                                                                                                                                                       | ≥ 90.8 %                                                                                                                                                                                                                                                       |
| HMMS (high molecular mass species) percentage by SE-HPLC | ≤ 6.5 %                                                                                                                                                                                                                                                        | ≤ 6.5 %                                                                                                                                                                                                                                                        |
| LMMS (low molecular mass species) percentage by SE-HPLC  | ≤ 4.5 %                                                                                                                                                                                                                                                        | ≤ 4.5 %                                                                                                                                                                                                                                                        |

# Timeline for Sciensano

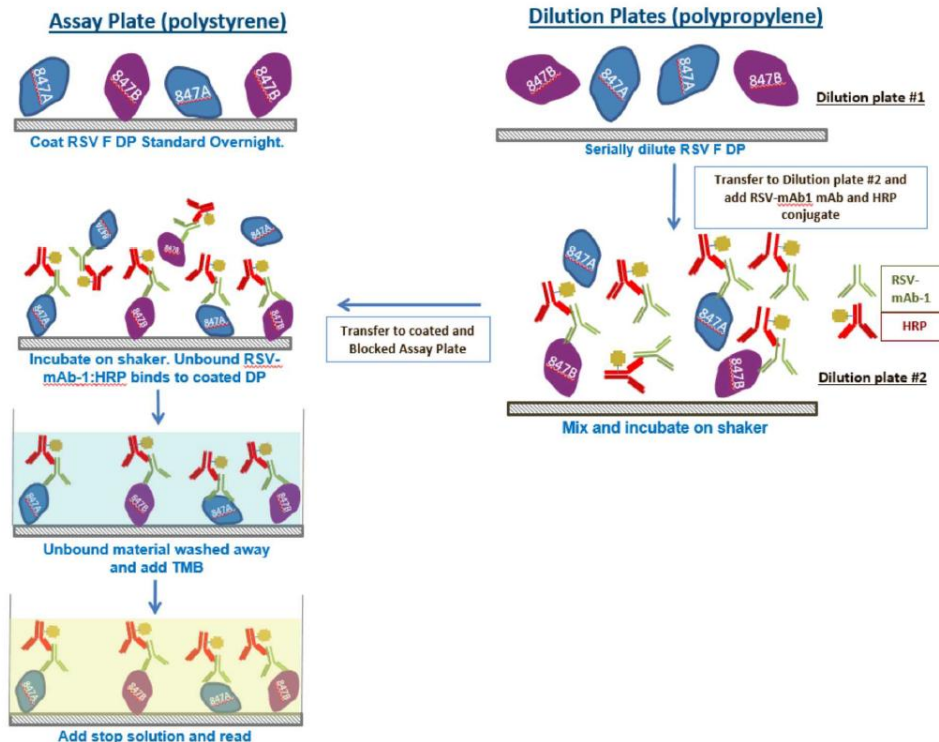
- Two OMCLs contacted by Pfizer for Abrysvo → Sciensano & ANSM



# RSVPreF *in vitro* relative potency by ELISA

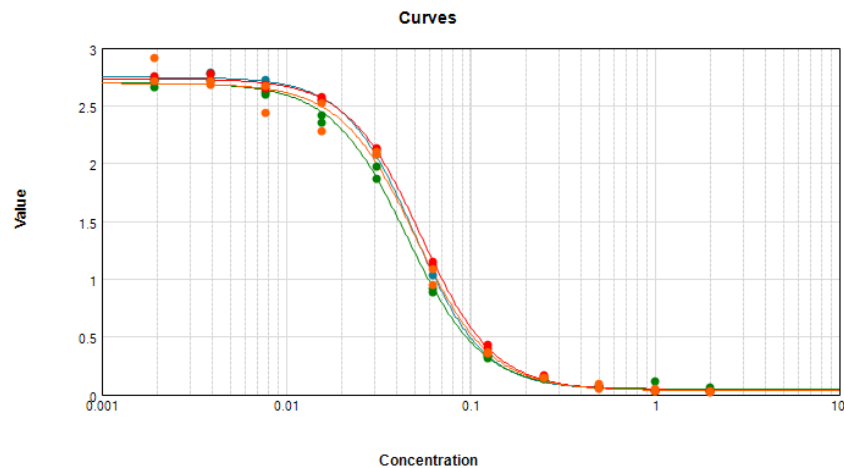
RSVPreF *in vitro* relative potency is measured by an inhibition ELISA:

- Dilution of samples and incubation with RSV-mAb and HRP-conjugated detection Ab
- Transfer on ELISA plate coated with RSV F DP standard (847A+847B)
- Coloration with TMB and reading



# RSVPreF *in vitro* relative potency by ELISA

- Results are expressed as % of Relative Potency against a RSVPreF qualified reference standard.
- $$\text{Relative efficacy (\%)} = \frac{EC50_{Std}}{EC50_{Sample}} \times 100$$



● **Standard** ( Standards@Plate 1: MeanValue vs Concentration ) Weighting: Fixed  
● **Control** ( Control@Plate 1: Values vs Concentration ) Weighting: Fixed  
● **Sample1** ( Sample1@Plate 1: Values vs Concentration ) Weighting: Fixed  
● **Sample2** ( Sample2@Plate 1: Values vs Concentration ) Weighting: Fixed

Curve Fit Results ▲

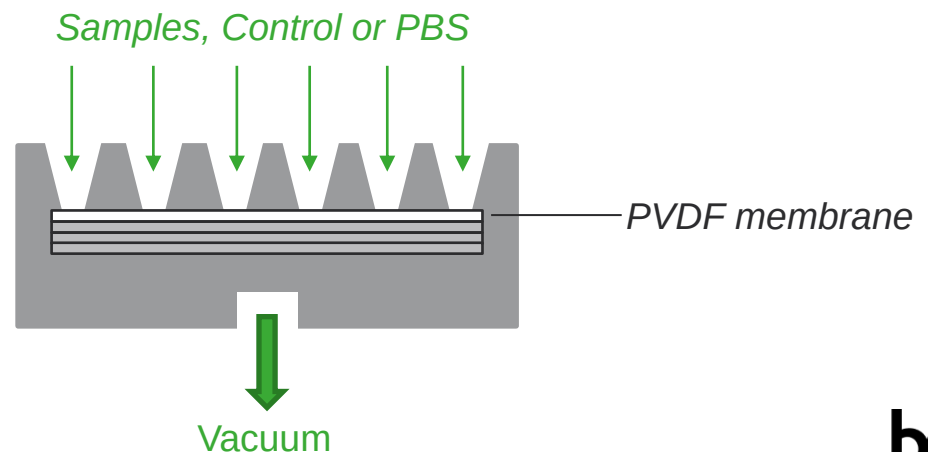
Curve Fit : 4-Parameter Logistic  $y = D + \frac{A - D}{1 + (\frac{x}{EC50})^B}$

|                                                           | Parameter | Estimated Value | Std. Error | Confidence Interval |
|-----------------------------------------------------------|-----------|-----------------|------------|---------------------|
| <b>Standard</b><br>R <sup>2</sup> = 1.000<br>EC50 = 0.050 | A         | 2.748           | 0.017      | [2.707, 2.790]      |
|                                                           | B         | 2.298           | 0.084      | [2.100, 2.495]      |
|                                                           | C         | 0.050           | 8.96e-4    | [0.047, 0.052]      |
|                                                           | D         | 0.045           | 0.016      | [0.008, 0.082]      |
| <b>Control</b><br>R <sup>2</sup> = 0.999<br>EC50 = 0.045  | A         | 2.697           | 0.021      | [2.652, 2.742]      |
|                                                           | B         | 2.128           | 0.090      | [1.939, 2.318]      |
|                                                           | C         | 0.045           | 0.001      | [0.043, 0.047]      |
|                                                           | D         | 0.048           | 0.018      | [0.010, 0.087]      |
| <b>Sample1</b><br>R <sup>2</sup> = 1.000<br>EC50 = 0.053  | A         | 2.735           | 0.011      | [2.712, 2.757]      |
|                                                           | B         | 2.189           | 0.048      | [2.087, 2.290]      |
|                                                           | C         | 0.053           | 6.09e-4    | [0.052, 0.055]      |
|                                                           | D         | 0.039           | 0.010      | [0.018, 0.060]      |
| <b>Sample2</b><br>R <sup>2</sup> = 0.994<br>EC50 = 0.050  | A         | 2.695           | 0.043      | [2.604, 2.785]      |
|                                                           | B         | 2.144           | 0.188      | [1.748, 2.540]      |
|                                                           | C         | 0.050           | 0.002      | [0.045, 0.055]      |
|                                                           | D         | 0.036           | 0.039      | [-0.045, 0.116]     |

# Identity by Slot Blot

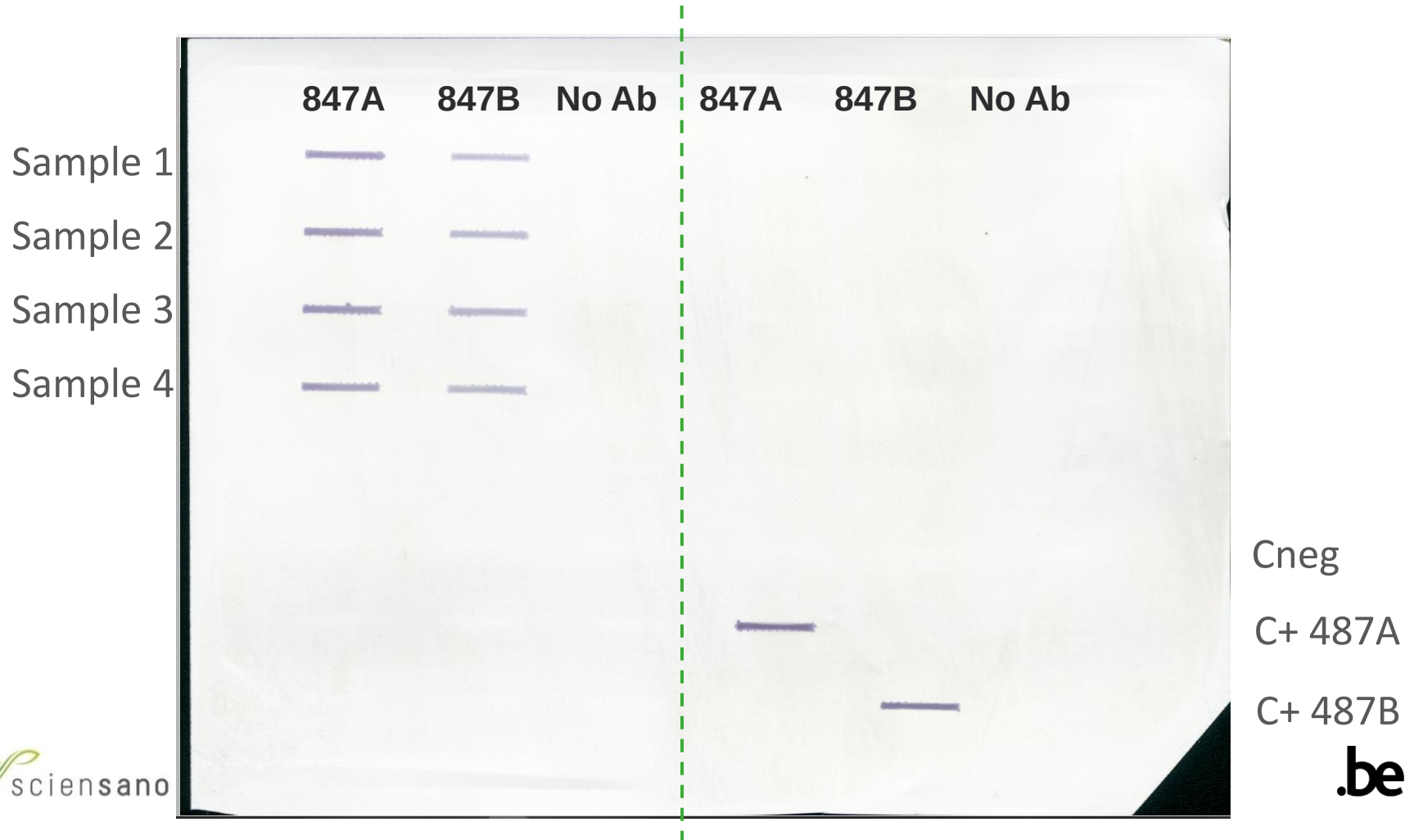
RSVPreF identity is determined by Slot Blot:

- Samples are diluted and adsorbed onto a PVDF membrane by vacuum filtration
- Membrane is incubated with specific mAb for RSVPreF 847A or 847B serotypes
- Membrane is then incubated with BSA to block the free-binding site
- Membrane is finally incubated with protein A/G HRP prior to coloration with 4CN to develop purple bands



# Identity by Slot Blot

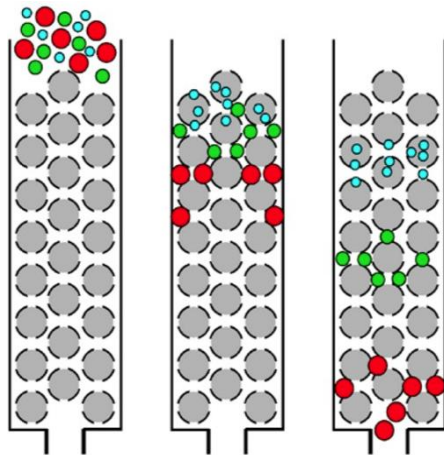
- Identity positive if a purple band develops





# Trimer, HMMS & LMMS percentage by SE-HPLC

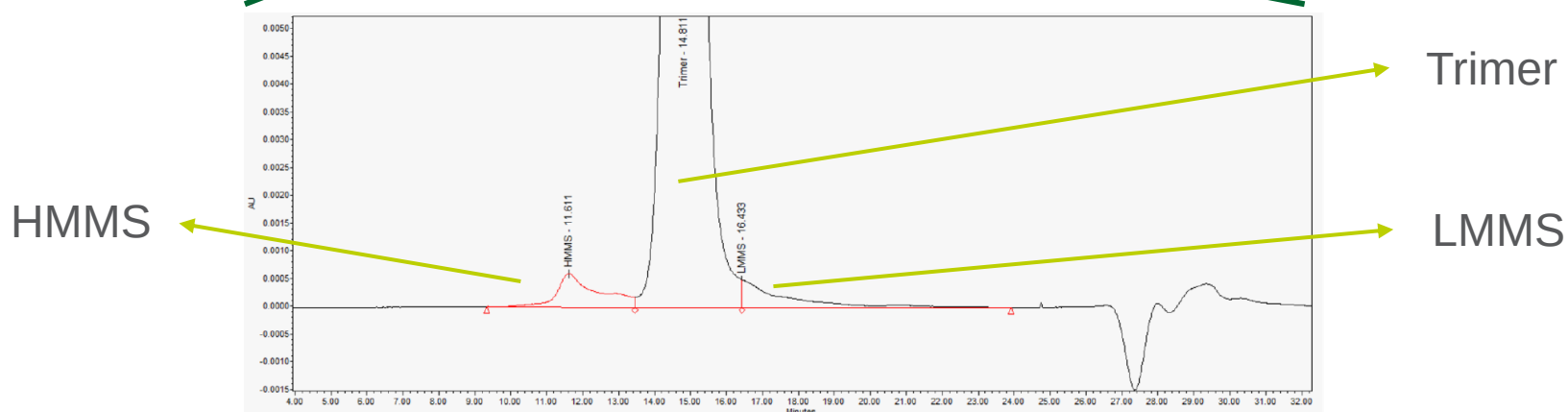
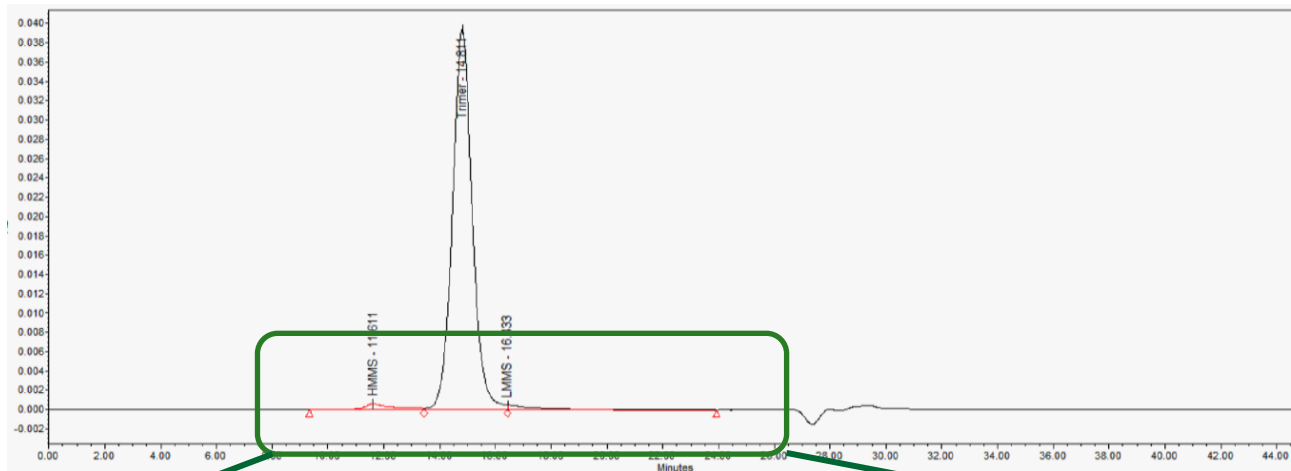
- **Size-exclusion chromatography - High performance liquid chromatography**
- Used to determine the relative content of :
  - HMMS (high molecular mass species)
  - Trimer (antigens)
  - LMMS (low molecular mass species)
- Separation of the molecules based on their hydrodynamic volume



*Custom Size-Exclusion Chromatography Service ([creative-biostructure.com](http://creative-biostructure.com))*

- Mobile phase salty (200 mM sodium chloride, 50 mM sodium phosphate, pH 7,4)
- Isocratic gradient, 100% mobile phase
- Detection = UV detector 280 nm

# Trimer, HMMS & LMMS percentage by SE-HPLC



% Area of trimer peak = measure of trimer purity

Specification: not less than 90.8%

# Method validation at Sciensano

Analytical parameters checked during method implementation

| ELISA                  |                   |
|------------------------|-------------------|
| Repeatability          | ✓                 |
| Intra-reproductibility | ✓                 |
| Inter-reproducibility  | ✓                 |
| parallelism            | ✓                 |
| Specificity            | ✓                 |
| Expanded uncertainty   | ✓                 |
| Robustness             | Not performed yet |

| Slot blot              |   |
|------------------------|---|
| Repeatability          | ✓ |
| Intra-reproductibility | ✓ |
| Selectivity            | ✓ |
| Specificity            | ✓ |

# Method validation at Sciensano

Analytical parameters checked during method implementation

| SE-HPLC               |                   |
|-----------------------|-------------------|
| Repeatability         | ✓                 |
| Intra-reproducibility | ✓                 |
| Inter-reproducibility | ✓                 |
| LOQ                   | ✓                 |
| Linearity             | ✓                 |
| Accuracy              | On going          |
| Specificity           | ✓                 |
| Expanded uncertainty  | ✓                 |
| Robustness            | Not performed yet |

# Challenges/lessons learned

- Very positive that two OMCLs are involved in this process
  - Allow exchange between the two OMCLs for method implementation (technical point of view)
  - Allow to be more « powerful » when requesting additional information from the manufacturer for the protocol
  
- Challenges :
  - Short timeline for method implementation → Method validation minimum, more data should be gathered afterwards
  - Convince the manufacturer to share some data for the protocol
  - In routine : protocol received before the samples – Pfizer explained that they send samples to OMCLs after receiving orders for EU market. Main market = USA

## Contact

Flore Laurent • [flore.laurent@sciensano.be](mailto:flore.laurent@sciensano.be) • + 32 2 642 5888

Julien Auquier • [julien.auquier@sciensano.be](mailto:julien.auquier@sciensano.be) • + 32 2 642 5700  
(Potency by ELISA and identity by Slot Blot)